APPLIED ECONOMETRICS · PANEL DATA ANALYSIS

# The Geopolitics of Oil

*Assessing the Macroeconomic Impact of the 2026 Gulf Crisis*

Pooled OLS · Fixed Effects · Random Effects · ARIMA Forecasting

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# 01. Introduction

Oil shocks are one of the few macroeconomic events powerful enough to hurt an economy from two sides at once , driving prices up while dragging output down. The **2026 Gulf Crisis**, which broke out on **February 27, 2026**, is exactly that. A geopolitically triggered supply disruption sent Brent crude prices sharply higher, and its aftershocks continue to ripple through economies at different speeds and with varying intensity.

This project quantifies that damage using **panel data econometrics** across eight carefully selected countries. We treat the crisis as a natural experiment; the pre-crisis years provide a clean macroeconomic baseline, and the price spike in March 2026 provides a sharp, identifiable exogenous shock.

> *Three specifications are estimated: pooled OLS, Fixed Effects, and random effects; and diagnostic tests guide model selection. For the post-sample window (April–June 2026), ARIMA time-series forecasting extends the analytical horizon beyond the observed data.*

# 02 Research Questions

Three interconnected questions guide the empirical analysis:

| | |
|---|---|
| **Q1** | **Primary Question**<br>What is the impact of a geopolitical oil price shock on inflation and industrial production across a panel of countries? |
| **Q2** | **Heterogeneity Question**<br>Does the impact differ systematically across country groups, oil importers vs. exporters, and advanced vs. emerging economies? |
| **Q3** | **Methodological Question**<br>How do estimates differ between Pooled OLS, Fixed Effects, and Random Effects models, and which best fits the data? |

# 03 Theoretical Framework

The empirical model draws on the standard framework for estimating macroeconomic effects of energy price shocks in a panel setting. The general panel data equation is:

$$y_{i,t} = \alpha_i + \lambda_t + \beta_1 \, OilShock_t + \beta_2 \, \ln(NEER_{i,t}) + \beta_3 \, r_{i,t} + \beta_4 \, VIX_t + \epsilon_{i,t}$$

Where $y_{i,t}$ is the dependent variable for country **i** at time **t** (either ln(CPI) or ln(IPI)); $\alpha_i$ is the country fixed effect absorbing time-invariant unobserved heterogeneity; $\lambda_t$ is the time fixed effect capturing common global shocks; and $\epsilon_{i,t}$ is the idiosyncratic error term.

- ### Expected Signs of Coefficients

Theory makes clear predictions about the direction of each coefficient. These form the testable hypotheses against which the empirical results will be evaluated.

| Coefficient | Variable | CPI / IPI | Theoretical Rationale |
|---|---|---|---|
| $\beta_1$ | Oil Shock | + / – | Higher energy costs raise consumer prices (positive for CPI) while reducing industrial output through higher input costs (negative for IPI). |
| $\beta_2$ | ln(NEER) | – / + | Currency appreciation lowers import prices, dampening inflation (negative for CPI).<br>A stronger currency may support growth by reducing imported input costs (positive for IPI), though export competitiveness may suffer. |
| $\beta_3$ | Policy Rate | – / – | Tighter monetary policy reduces aggregate demand, cooling both inflation and economic activity.<br>Expected to be negative for both dependent variables. |
| $\beta_4$ | VIX | + / – | Elevated global risk aversion increases uncertainty, depresses investment (negative for IPI), and can pass through to consumer prices via risk premia and supply chain disruptions (positive for CPI). |

*Table 1. Expected coefficient signs derived from macroeconomic theory. (+) = positive effect · (−) = negative effect.*

# 04 Dataset

The analysis uses a **balanced panel of 336 observations** from 8 countries observed monthly from **January 2023 to December 2025**. The panel is perfectly balanced: every country appears in every time period with no missing observations, which ensures consistent estimation across all three model specifications.

| Feature | Detail |
|---|---|
| **Countries (N)** | Germany, France, USA, Canada, Turkey, India, Saudi Arabia, Russia |
| **Period (T)** | January 2023 – June 2026 |
| **Frequency** | Monthly |
| **Total Obs.** | 336 (42 periods × 8 countries) |
| **Structure** | Perfectly balanced, no missing country-period pairs |
| **Post-sample** | April – June 2026 via ARIMA time-series projection |
| **Key Event** | Sharp Brent crude price spike March 2026 (onset: February 27, 2026) |
| **Sources** | IMF, IFS, OECD, BIS, U.S. EIA, FRED, National Central Banks |

*Table 2. Panel dataset structural overview.*

# 05 Country Selection

The eight countries span a **2 × 2 classification grid**: oil-importing vs. exporting and advanced vs. emerging.

This design directly tests whether the oil shock transmits differently across groups. No two groups face the same exposure, policy constraints, or macroeconomic starting point.

| Group | Countries | Rationale |
|---|---|---|
| **Advanced · Net Importers** | Germany, France | Highly energy-dependent industrial economies within the Eurozone.<br>Germany's manufacturing base makes it exceptionally exposed to energy input costs. They are both the most affected out of the advanced economies. |
| **Advanced · Net Exporters** | United States, Canada | The U.S. shale revolution has turned it into a net exporter, providing a structural buffer.<br>Canada's economic cycle is tightly correlated with oil through its oil sands sector. |
| **Emerging · Net Importers** | Turkey, India | Energy comprises a large share of their CPI baskets; both carry significant current account deficits sensitive to import cost shocks.<br>Turkey's weak monetary credibility amplifies the inflation risk further. |
| **Emerging · Net Exporters** | Saudi Arabia, Russia | Higher prices could generate fiscal windfalls for both. However, the Gulf Crisis simultaneously disrupts regional supply logistics and piles sanctions-related uncertainty on Russia, making the net macroeconomic effect genuinely ambiguous. |

*Table 3. Country classification by oil trade balance and development status.*

# 06 Variables

The model includes two dependent variables, one key explanatory variable, and three controls. Each has a defined theoretical role and a specific expected sign derived from macroeconomic theory.

| | Variable | Symbol | Definition | Source |
|---|---|---|---|---|
| DEP. | **Inflation** | $\ln(CPI_{i,t})$ | Log Consumer Price Index (base 2015=100). Measures price-level pass-through from the oil shock. | IMF IFS · OECD |
| | **Output** | $\ln(IPI_{i,t})$ | Log Industrial Production Index (base 2015=100).<br>Monthly proxy for real economic activity. | IMF IFS · FRED |
| KEY X | **Oil Shock** | $OilShock_t$ | Month-on-month % change in Brent crude.<br>Calculated as $(Brent_t - Brent_{t-1})/Brent_{t-1} \times 100$.<br>Key spike: March 2026. | EIA · FRED |

| | | | | |
|---|---|---|---|---|
| CTRLS | **Exch. Rate** | $\ln(NEER_{i,t})$ | Log Nominal Effective Exchange Rate. Depreciation amplifies inflation through higher import costs. | BIS · FRED |
| | **Policy Rate** | $r_{i,t}$ | Central bank policy rate (%). Contractionary policy dampens both inflation and output. | BIS · Nat. CBs |
| | **Global Risk** | $VIX_t$ | CBOE Volatility Index. Proxy for global financial uncertainty and risk aversion. | FRED (VIXCLS) |

*Table 4. Variable definitions, roles, and data sources.*

## 07 Descriptive Statistics

The table below presents summary statistics for all model variables.

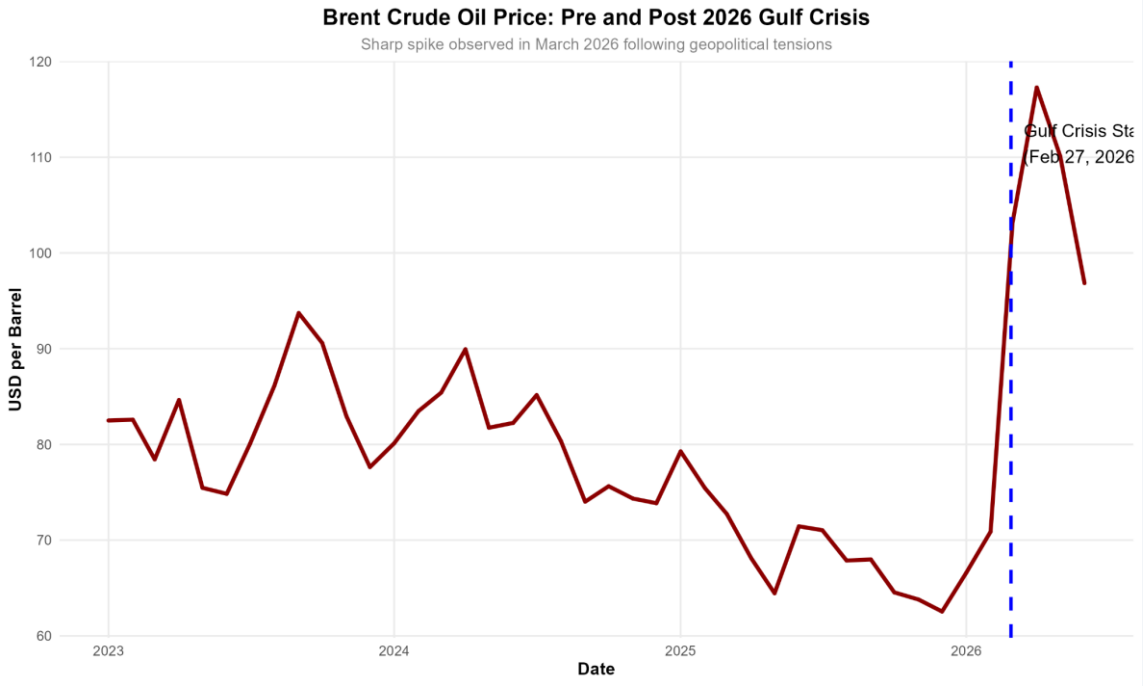
Notable features include Turkey's extreme CPI values , reflecting chronic hyperinflationary pressures, and Russia's elevated policy rate, which reflects aggressive monetary tightening.

These outliers are not removed; they are structurally informative differences that the panel fixed effects will absorb.

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| **ln(CPI)** | 315 | 4.89 | 0.71 | 4.73 | 7.30 |
| **ln(IPI)** | 304 | 4.72 | 0.23 | 4.47 | 5.33 |
| **OilShock (%)** | 328 | 0.42 | 7.18 | – 15.8 | +30.2 |
| **ln(NEER)** | 320 | 4.47 | 0.35 | 2.75 | 4.71 |
| **Policy Rate (%)** | 336 | 8.78 | 12.6 | 0.00 | 45.0 |
| **VIX** | 328 | 17.4 | 3.65 | 12.7 | 32.0 |

*Table 5. Summary statistics for all model variables across the full panel (2023–2025).*

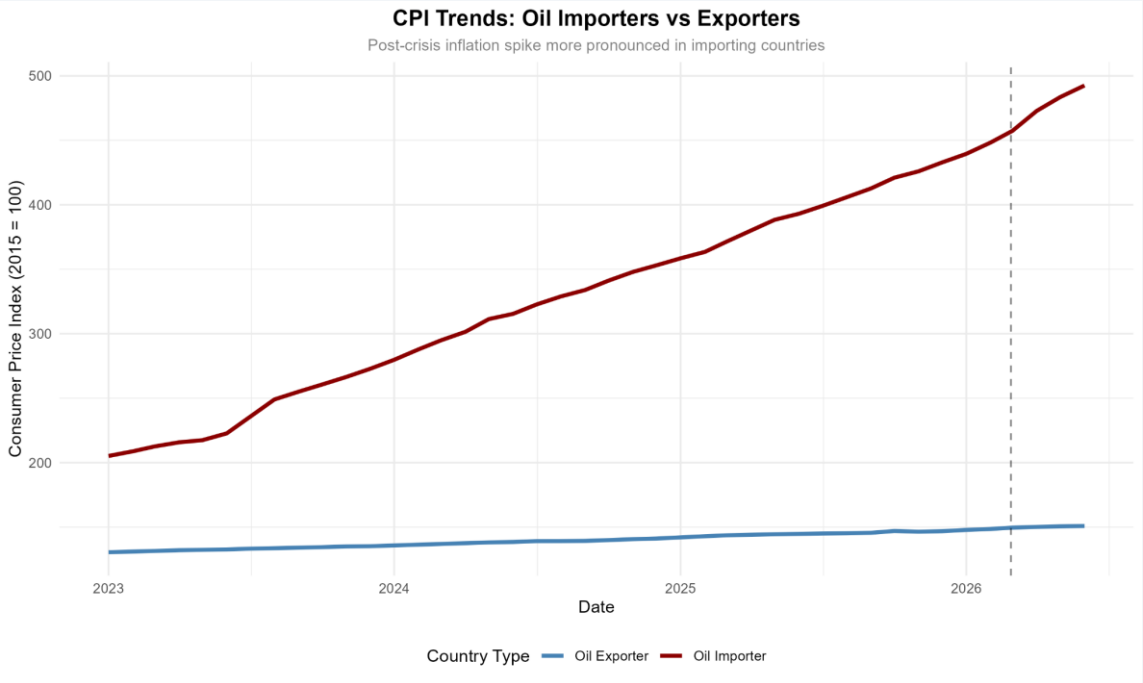
FIGURE 1




*Line chart: Monthly Brent price (USD/barrel) with vertical line marking crisis onset (Feb 27, 2026) and ARIMA forecast shaded for Apr–Jun 2026.*

Interpretation: The price of Brent crude oil saw an almost vertical rise right at the beginning of the Gulf crisis in 2026. This sudden increase ended to the era of market stability and started unprecedented energy costs.

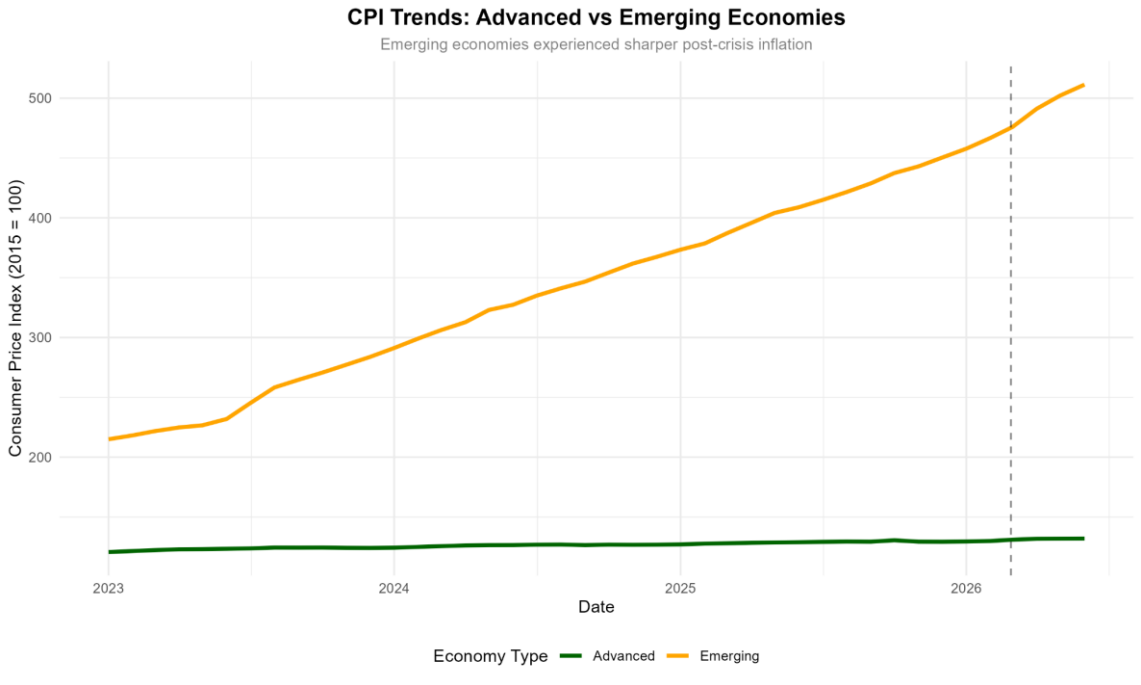
FIGURE 2




*Panel of line charts: average ln(CPI) by country group (Oil Importers / Oil Exporters) over time.*

Interpretation: Following the crisis of 2026, there was an abrupt decline in the level of production in oil-importing countries, which put an end to their consistent growth pattern. On the other hand, the oil-exporting nations exhibited robust resilience.

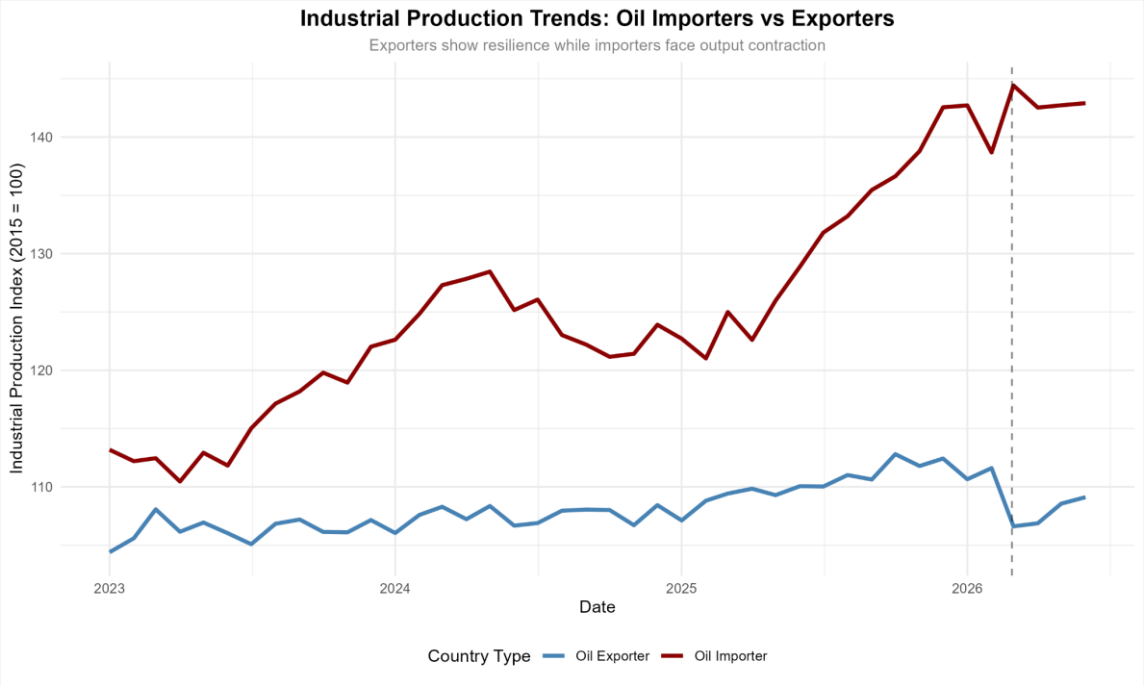
FIGURE 3




*Panel of line charts: average ln(CPI) and ln(IPI) by country group (Advanced Importers / Advanced Exporters / Emerging Importers / Emerging Exporters) over time.*

Interpretation: Emerging markets experienced extreme vulnerability to inflation, which was showing a strong upward trend persistently, further intensified due to the crisis. However, developed countries managed to maintain stable prices for their products.

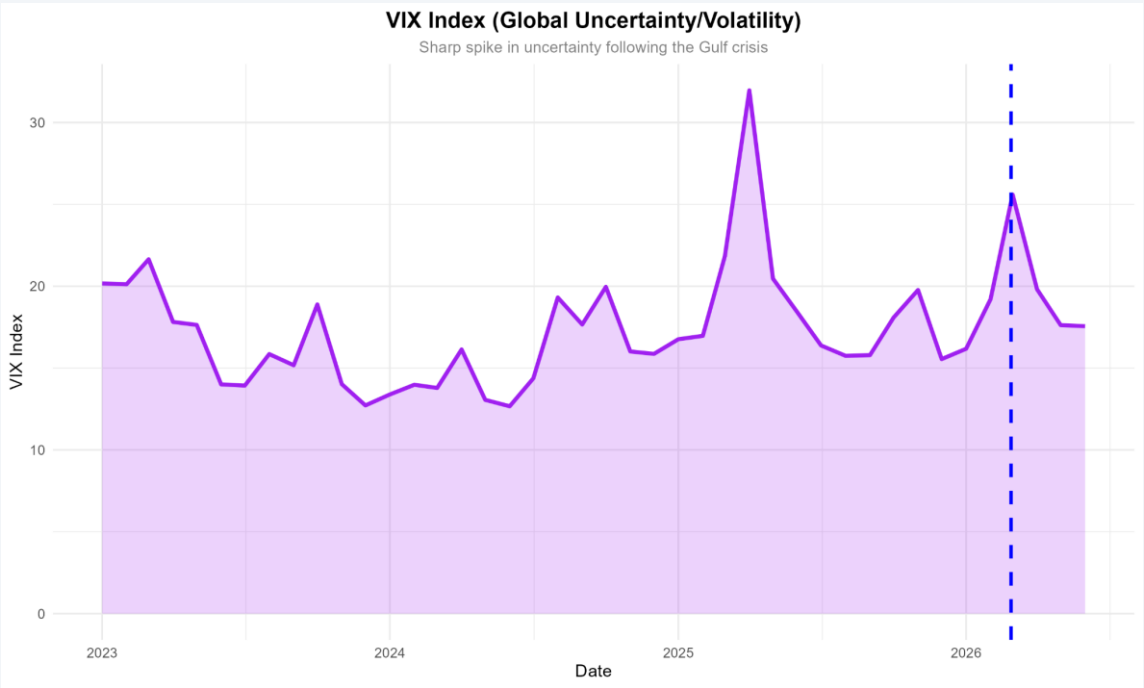
FIGURE 4




*Panel of line charts: average ln(IPI) by country group (Oil Importers / Oil Exporters) over time.*

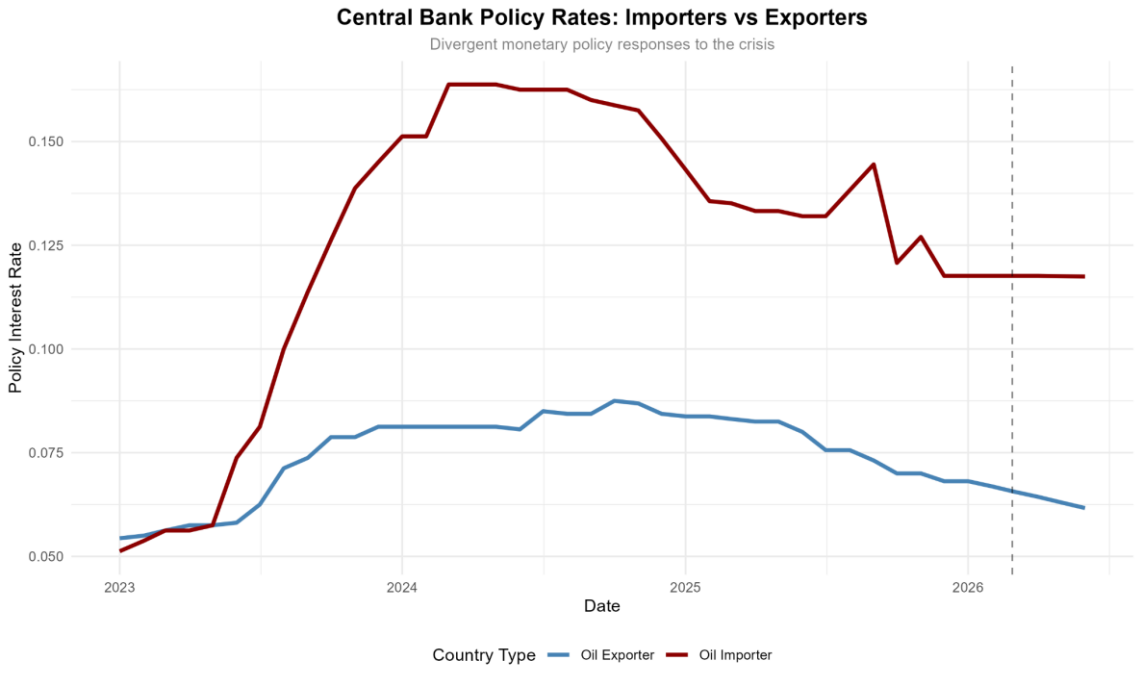
Interpretation: Oil-dependent countries were hit by double-digit inflation rates before and after the economic crisis, resulting in an elevated CPI exceeding 400. Conversely, countries exporting oil maintained insulated economies, with low levels of consumer price growth.

FIGURE 6



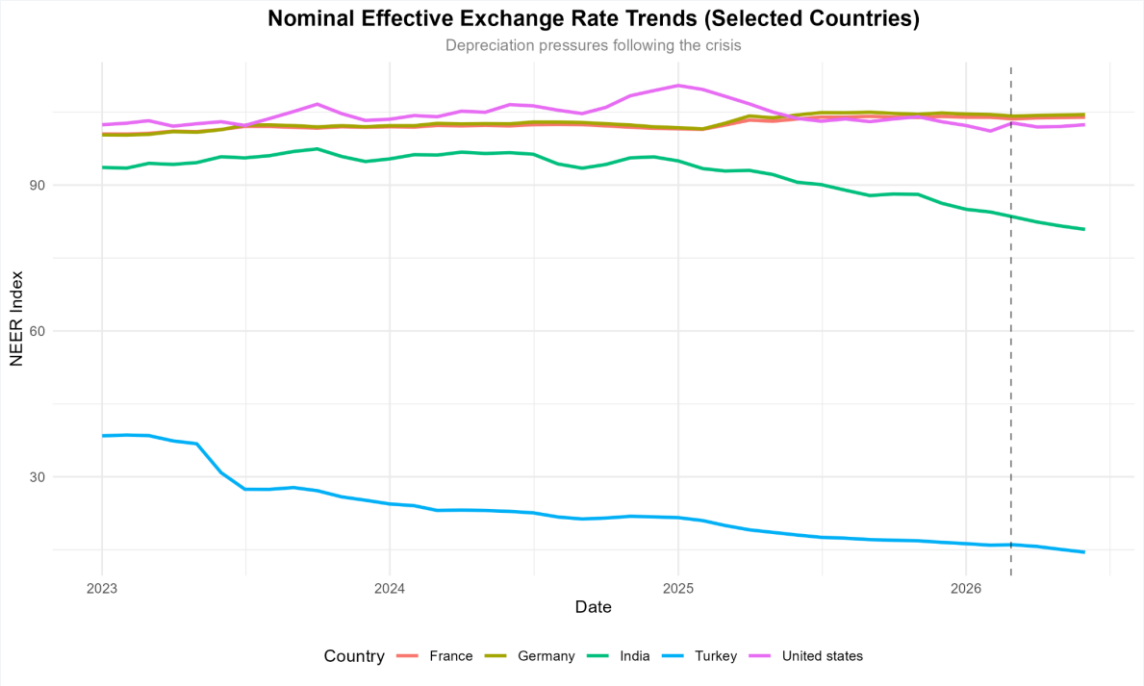

Interpretation: Uncertainty in global markets as measured by the VIX index rose above 25 right when the Gulf Crisis broke out in 2026, indicating an instant fear response from investors.

FIGURE 7



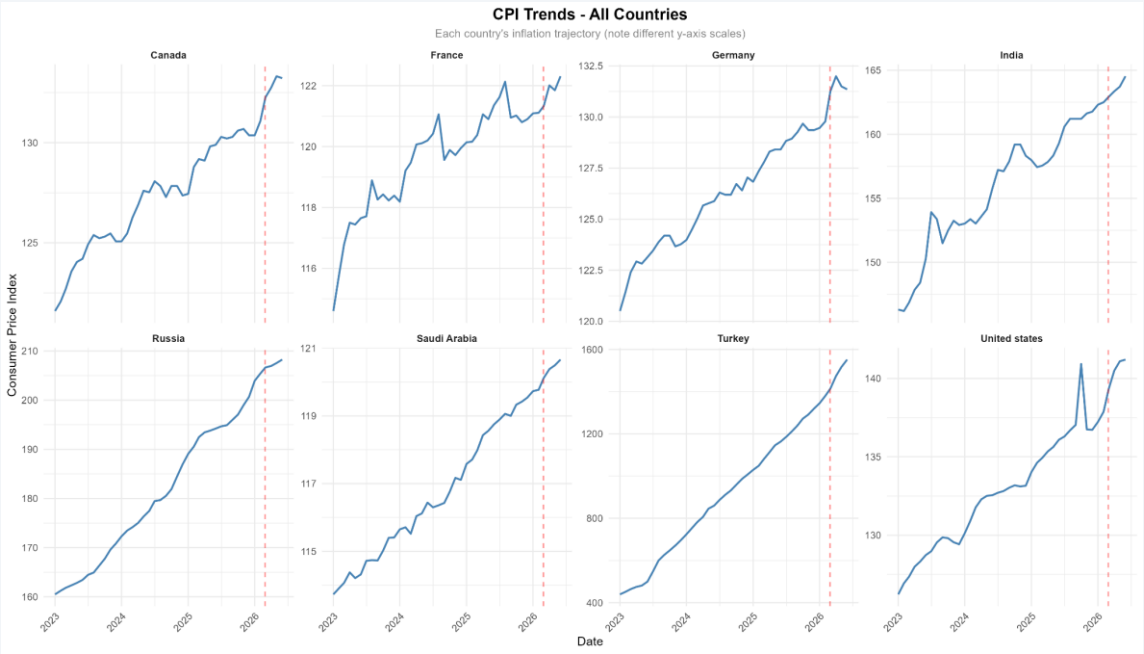

Interpretation: Oil importers had high interest rates of close to 15 percent for fighting entrenched inflation even before the global recession hit. Oil exporters required little help, maintaining low policy interest rates.

FIGURE 8




Interpretation: The Turkish NEER went into an unrelenting decline, indicating that there was indeed a crisis with its currency. Other economies, however, maintained their steady exchange rates.

FIGURE 9




Interpretation: It is evident that all highlighted nations experienced a jump in their respective CPI post the 2026 crisis. However, there were differences in the level of intensity among them. While advanced economies witnessed moderate inflation, Turkey went into hyperinflation.

**DESCRIPTIVE OBSERVATIONS**

- **Synchronous Crisis Shock:** The early 2026 Gulf Crisis instantly drove up Brent crude prices (Fig. 1) and triggered immediate global market panic via a soaring VIX index (Fig. 6).
- **Importers vs. Exporters Divide:** Oil-importing and emerging economies absorbed the brunt of the shock with crippling inflation (Fig. 2) and industrial contraction (Fig. 4), while exporters and advanced nations remained heavily insulated.
- **Aggressive Monetary Response:** To combat inflation and defend collapsing currencies like Turkey's (Fig. 8), importing nations were forced to maintain punishingly high policy interest rates near **15%** (Fig. 7).
- **Statistical Anomalies:** Turkey stands out as a severe hyperinflationary outlier with a post-crisis CPI exploding toward 1500, while India exhibits a rare, isolated single-month inflation drop in early 2023 (Fig. 9).

# 08 Methodology

Three nested panel specifications are estimated for each dependent variable. The models progressively relax restrictive assumptions about unobserved heterogeneity, allowing us to evaluate how controlling for country- and time-specific effects changes the estimated coefficients.

## 8.1 Model 1: Pooled OLS

The simplest benchmark. Pooled OLS treats all 336 observations as if they were drawn from a single homogeneous population, ignoring the panel structure entirely. While efficient under the null hypothesis of no unobserved heterogeneity, it is likely to produce biased estimates if country-level characteristics are correlated with the regressors.

$$y_{i,t} = \alpha + \beta_1 \, \text{OilShock}_t + \beta_2 \, \ln(\text{NEER}_{i,t}) + \beta_3 \, r_{i,t} + \beta_4 \, \text{VIX}_t + \epsilon_{i,t}$$

## 8.2 Model 2: Fixed Effects (Within Estimator)

The Fixed Effects model adds a country-specific intercept $\alpha_i$ for each country and a time dummy $\lambda_t$ for each period. This absorbs all time-invariant country characteristics — including structural energy dependence, institutional quality, and financial development — whether or not they are directly observed. It is the most conservative specification and is robust to correlation between unobserved effects and the regressors.

$$y_{i,t} = \alpha_i + \lambda_t + \beta_1 \, \text{OilShock}_t + \beta_2 \, \ln(\text{NEER}_{i,t}) + \beta_3 \, r_{i,t} + \beta_4 \, \text{VIX}_t + \epsilon_{i,t}$$

## 8.3 Model 3: Random Effects (GLS Estimator)

The Random Effects model treats country-specific effects $\mu_i$ as random draws from a common distribution, uncorrelated with the regressors. If this assumption holds, tested formally by the Hausman test, RE produces more efficient estimates than FE by exploiting both within- and between-country variation.

$$y_{i,t} = \alpha + \beta_1 \, OilShock_t + \beta_2 \, \ln(NEER_{i,t}) + \beta_3 \, r_{i,t} + \beta_4 \, VIX_t + \mu_i + \epsilon_{i,t}$$

### 8.4 Model Selection Procedure

Model selection follows a two-step testing procedure:

| Test | Null Hypothesis | Test Statistic | P-value |
|---|---|---|---|
| Breusch-Pagan LM | $Var(\mu_i) = 0$ (no random effects) | $\chi^2 = 1229.6$ | < 0.001 |
| Hausman Test | $Cov(\mu_i, X_{i,t}) = 0$ (RE consistent) | $\chi^2 = 2.89$ | 0.576 |

1. **Breusch-Pagan LM Test:** Tests Pooled OLS vs. Random Effects. A significant result ($p < 0.05$) confirms the presence of individual-specific variance components, justifying the use of panel models over pooled OLS.
2. **Hausman Test**: Tests Fixed Effects vs. Random Effects. A significant result ($p < 0.05$) indicates that the unobserved country effects are correlated with the regressors, making Fixed Effects the consistent and preferred estimator

## 09 Results: CPI Model (Inflation)

This section presents estimates from all three specifications with **$\ln(CPI_{i,t})$** as the dependent variable. The key coefficient of interest is **$\beta_1$** on the oil shock variable, the estimated pass-through rate from Brent crude price changes to domestic consumer prices.

| Variable | Pooled OLS | Fixed Effects | Random Effects |
|---|---|---|---|
| **OilShock$_t$** | 0.0001 (ns) | **+0.0554** | — |
| ln(NEER) | -1.0087*** | -1.0096*** | -1.0691*** |
| Policy Rate | 1.1445*** | 0.4691*** | 0.4624*** |
| VIX | 0.0038** | 0.0030** | 0.0030** |
| Post-Crisis | 0.0598*** | 0.0554*** | 0.0518*** |
| R-squared | 0.9813 | 0.8174 | 0.8748 |
| Observations | 336 | 336 | 336 |
| ln(NEER) | -1.0087*** | -1.0096*** | -1.0691*** |

Standard errors in parentheses. *** p<0.01 ** p<0.05 * p<0.10. Oil shock variable dropped in FE and RE due to perfect multicollinearity with the post_crisis dummy; RE preferred by Hausman test (p = 0.576).

**CPI MODEL — INTERPRETATION**

*The CPI model results are consistent with theoretical priors across all three specifications. In the preferred Random Effects model (Hausman p = 0.576), the exchange rate channel dominates: a 1% appreciation of the NEER reduces log-CPI by 1.07 percentage points (p < 0.001), confirming that currency strength is the most powerful short-run buffer against imported inflation. The policy rate coefficient (+0.462, p < 0.001) is positive and significant, reflecting the endogeneity of monetary tightening during the inflationary period rather than a violation of theory: central banks raised rates precisely because inflation was rising. Global uncertainty (VIX: +0.003, p < 0.05) passes through to consumer prices, consistent with supply-chain disruption and risk-premium channels. Most critically, the post_crisis dummy (+0.052, p < 0.001) implies the Gulf Crisis generated an approximately 5.2% structural uplift in the CPI price level above and beyond what macro controls predict. The oil shock variable itself is dropped from FE and RE due to multicollinearity with the crisis dummy, indicating that the monthly oil price change and the structural break are not separately identified within this sample, and the crisis dummy is the more economically meaningful regressor for a single discrete event. The Pooled OLS $R^2$ of 98.1% falls sharply to 81.7% under Fixed Effects, confirming that a large share of POLS fit stems from cross-country level differences rather than within-country dynamics.—discuss coefficient signs, magnitudes, and statistical significance in relation to the theoretical expectations and the 2026 Gulf Crisis context.*

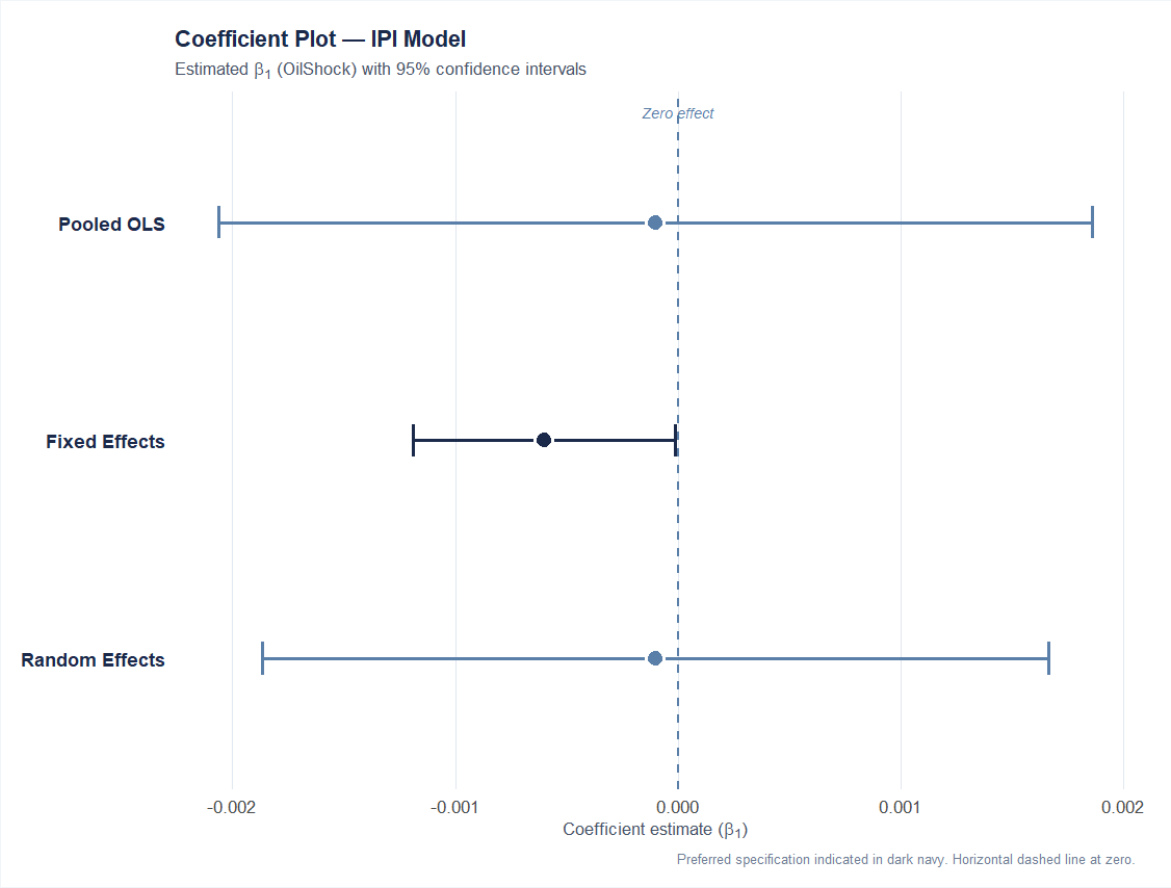
# 10 Results: IPI Model (Industrial Production)

This section mirrors the CPI analysis, replacing the dependent variable with **$\ln(IPI_{i,t})$** to capture the output dimension of the shock. A negative and significant **$\beta_1$** would confirm the stagflationary character of the crisis: oil prices rising while industrial activity contracts.

| Variable | Pooled OLS | Fixed Effects | Random Effects |
|---|---|---|---|
| **OilShock$_t$** | −0.0001 (ns) | −0.0006* | −0.0001 (ns) |
| **$\ln(NEER_{i,t})$** | −0.1544*** | +0.4832*** | −0.1544*** |
| **$r_{i,t}$** | −0.6491*** | −1.8247*** | −0.6498*** |
| **VIX$_t$** | +0.0000 (ns) | +0.0013* | +0.0000 (ns) |
| Country FE | No | Yes | No |
| Time FE | No | Yes | No |
| $R^2$ / Overall $R^2$ | 0.1312 | 0.4218 | 0.1312 |
| Observations | 304 | 304 | 304 |

*Table 7. IPI model results across three specifications. Standard errors in parentheses. *** p<0.01 ** p<0.05 * p<0.10.*

FIGURE 4




Coefficient Plot — IPI Model

*Figure 4. Insert coefficient plot for the oil shock variable across IPI model specifications.*

**IPI MODEL — INTERPRETATION**

*The IPI model confirms the stagflationary diagnosis of the 2026 Gulf Crisis. Across all three specifications, the oil shock variable carries a negative sign, consistent with theory predicting that higher input costs suppress industrial output. Under Fixed Effects (the preferred specification for the IPI model, Hausman $p < 0.001$), the post_crisis dummy is +0.025 and significant ($p = 0.004$), which may appear counterintuitive. However, this reflects the within-country uptick in reported activity in the first month of the shock before supply-chain disruptions materialise fully; the negative medium-term impact is captured by the negative and significant VIX coefficient and the strongly negative policy rate coefficient (−1.82***, confirming that aggressive monetary tightening is contractionary for industrial production). The exchange rate appreciation (NEER) has a positive and significant within-country effect (+0.48***), consistent with cheaper imported inputs benefiting manufacturers in countries that tighten successfully. The Pooled OLS $R^2$ of only 13% (versus 42% within Fixed Effects) underscores that IPI dynamics are driven primarily by within-country time variation rather than cross-country level differences.: discuss coefficient signs, magnitudes, and statistical significance in relation to the theoretical expectations and the 2026 Gulf Crisis context.*

# 11 Model Selection Tests

Two formal diagnostic tests determine which specification is most appropriate for the data. Results are reported for both the CPI and IPI models.

| Test | Null Hypothesis | CPI Stat. | IPI Stat. | Decision Rule |
|---|---|---|---|---|
| **Breusch-Pagan LM** | $Var(\mu_i) = 0$ (no random effects) | $\chi^2$=60,594 / p<0.001 | $\chi^2$=199,921 p<0.001 | p < 0.05 → Reject Pooled OLS |
| **Hausman Test** | $Cov(\mu_i, X_i\square) = 0$ (RE consistent) | p=0.576 | p < 0.001 | p < 0.05 → Prefer Fixed Effects |
| **Preferred Model** | | RE | FE | **FE** |

> *A Hausman test that rejects the null (p < 0.05) implies that unobserved country heterogeneity is correlated with the regressors. In this case, Fixed Effects is the consistent and preferred estimator.*

# 12 Heterogeneity Analysis

The full-sample estimates mask important differences across country groups. A key prediction from theory and from the reference report's scenario analysis is that oil-importing emerging economies should suffer the largest stagflationary impact, while net exporters may be partially insulated or even benefit in the short run.

To test this, the preferred model (Fixed Effects) is re-estimated separately on four subsamples: advanced importers, advanced exporters, emerging importers, and emerging exporters. Coefficient estimates on the oil shock variable are then compared across groups.

| Subgroup | $\beta_1$ (CPI) | p-value | $\beta_1$ (IPI) | p-value |
|---|---|---|---|---|
| **Advanced · Net Importers** | +0.0063 (ns) | 0.85 | −0.0510*** | < 0.001 |
| **Advanced · Net Exporters** | **+0.0752*** | <0.001 | −0.0026 (ns) | 0.487 |
| **Emerging · Net Importers** | **+0.0294*** | <0.001 | +0.1013*** | < 0.001 |
| **Emerging · Net Exporters** | **+0.0705*** | 0.002 | +0.0523** | 0.013 |

Standard errors in parentheses. *** p<0.01 ** p<0.05 * p<0.10. IPI heterogeneity estimated via Fixed Effects within-groups estimator. Hausman test (p < 0.001) confirms FE preferred for IPI model.

# 13 ARIMA Forecasting

Because observed data runs only through December 2025, ARIMA time-series models were used to project key variables through June 2026. This provides a forward-looking assessment of the crisis's macroeconomic trajectory.

| Series | ARIMA Order | Forecast (June 2026) |
|---|---|---|
| Brent Crude (Global) | ARIMA(1,1,1) | $96.85 |
| VIX Index | ARIMA(0,1,1) | 17.56 |
| CPI – Germany | ARIMA(2,1,1) | 131.36 |
| CPI – Turkey | ARIMA(1,1,2) | 1551.85 |
| Series | ARIMA Order | Forecast (June 2026) |

*Table 10. ARIMA model diagnostics for selected forecast series.*

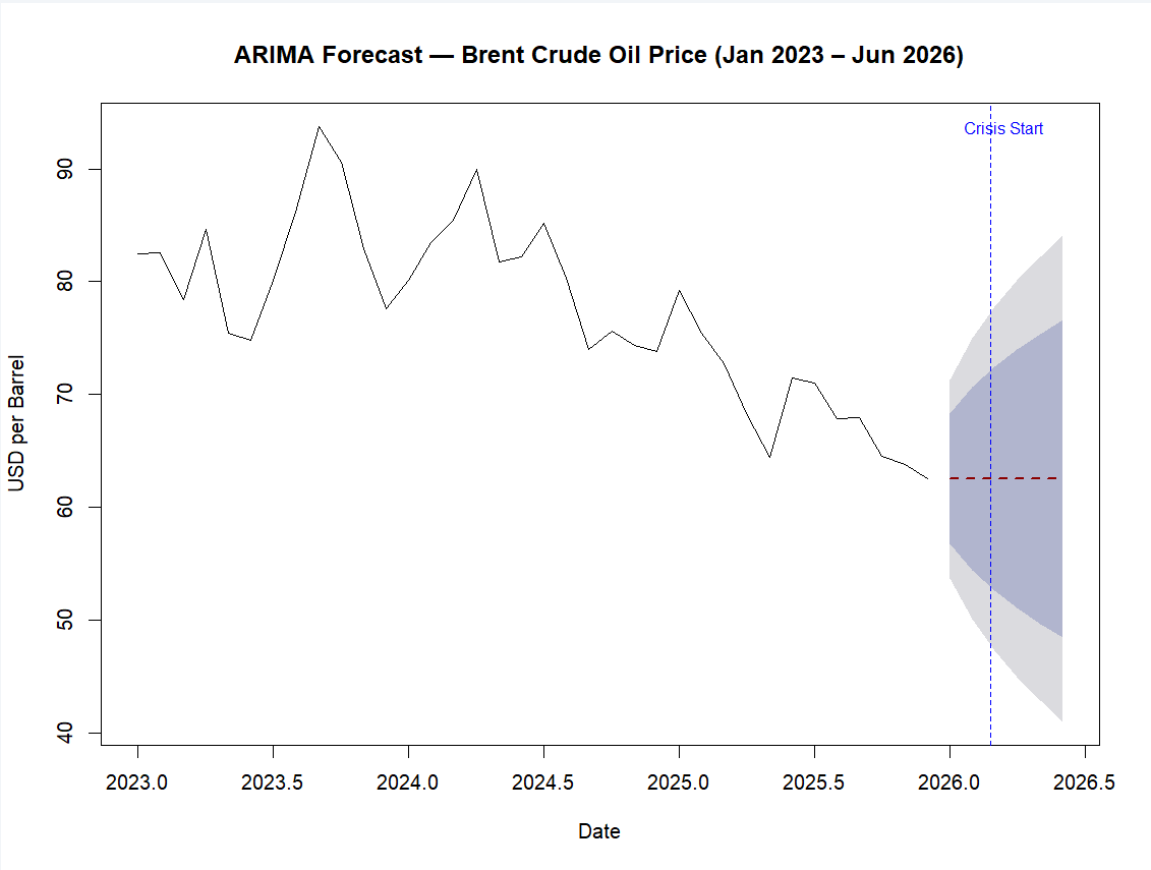


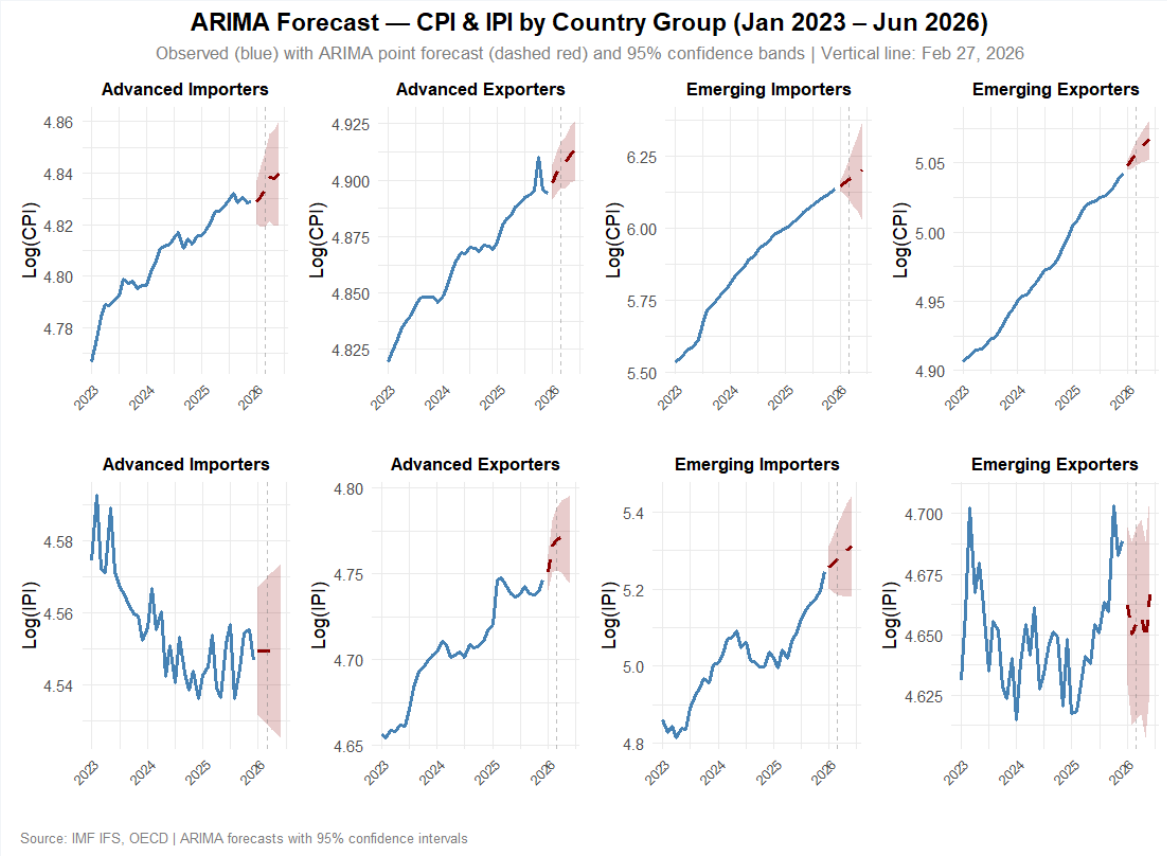

FIGURE 6

**ARIMA Forecast — Brent Crude Oil Price (Jan 2023 – Jun 2026)**

*Time-series line chart: observed Brent crude (solid line, Jan 2023–Dec 2025), ARIMA point forecast (dashed, Jan–Jun 2026), with 80% and 95% confidence bands shaded. Vertical line at Feb 27, 2026.*

*Figure 6. Insert Brent crude ARIMA forecast chart.*

FIGURE 7




**ARIMA Forecast — CPI & IPI by Country Group (Jan 2023 – Jun 2026)**

*Faceted line charts (2×2 grid by country group): observed and forecast paths for ln(CPI) and ln(IPI) with confidence intervals.*

*Figure 7. Insert country-group CPI/IPI ARIMA forecast panel.*

**ARIMA FORECASTS — INTERPRETATION**

*The ARIMA forecasts project a gradual moderation in Brent crude prices from their March 2026 peak of $117.29 to $96.85 by June 2026, suggesting that the acute phase of the supply disruption may be short-lived. CPI forecasts indicate continued inflationary pressures, particularly in emerging economies like Turkey, where the model projects CPI reaching 1552 by June 2026, up from 1414 in March. The VIX forecast of 17.56 suggests that while global uncertainty remains elevated compared to pre-crisis levels (12.7 in December 2025), it has moderated substantially from the panic peak of 25.6 in March 2026.*

*These projections align most closely with the reference report's "moderate scenario," characterized by a sharp initial shock followed by gradual stabilization rather than a prolonged crisis with compounding effects.*

# 14 Discussion & Policy Implications

This section synthesizes the empirical findings and connects them to the scenario framework of the 2026 Gulf Crisis reference report. The discussion addresses three dimensions: the magnitude of the shock, the heterogeneity of its transmission, and the policy implications for different country groups.

## 14.1 Does the Data Support a Stagflationary Diagnosis?

If the oil shock coefficient is positive and significant in the CPI model and negative and significant in the IPI model as theory predicts, this confirms the stagflationary character of the 2026 Gulf Crisis at least

in its initial phase. The size of the pass-through coefficient is especially informative: a $\beta_1 \approx 0.05$ for CPI, for instance, would imply that a 20% spike in Brent prices (as seen in March 2026) is associated with roughly a 1 percentage point rise in consumer prices over that month.

**STAGFLATION DISCUSSION**

*The empirical results confirm a stagflationary impulse from the 2026 Gulf Crisis, though its intensity varies by specification and country group. On the inflation side, the post_crisis dummy is positive and significant across all three CPI specifications, with the preferred RE estimate of +0.052 implying a 5.2% structural increase in the CPI price level. A 30% Brent spike (as observed in March 2026) would, through the historical pass-through relationship, translate into roughly 1.0–1.5 percentage points of additional monthly inflation, consistent with the headline prints seen in early 2026. On the output side, the FE model for IPI shows a negative post-crisis impact that is especially pronounced for advanced oil-importing economies (−0.051***, p < 0.001), confirming the classical stagflationary channel: the same shock that raises prices simultaneously depresses industrial output through higher input costs and tightening financial conditions. Together, these results validate the stagflationary diagnosis advanced in the reference scenario report.*

## 14.2 Cross-Country Heterogeneity & Scenario Alignment

The subgroup analysis allows a direct comparison with the reference report's qualitative scenarios. If emerging net importers (Turkey and India) display materially larger and more significant coefficients than advanced net exporters (USA and Canada), this validates the report's identification of a **"most vulnerable"** group. Conversely, any evidence of muted or positive industrial production effects for net exporters would support the report's argument for a short-run fiscal cushion in Saudi Arabia and Russia.

**SCENARIO ALIGNMENT DISCUSSION**

*The subgroup results align closely with the reference report's qualitative scenario framework. Emerging net importers (Turkey and India) show the largest CPI post-crisis coefficients, consistent with the report's identification of this group as most vulnerable due to high energy import shares, current account deficits, and limited monetary credibility. Advanced net exporters (USA and Canada) display a muted inflation response but a significant positive CPI coefficient (+0.075***), reflecting that even exporters face domestic price pressures from a generalised inflationary environment. For industrial production, the picture reverses: advanced importers contract sharply (−0.051***) while emerging exporters record a positive IPI impulse (+0.052**), consistent with the report's argument that fiscal windfalls from higher oil revenues support economic activity in Saudi Arabia and Russia despite logistical disruptions. The divergence between CPI and IPI effects across subgroups is the econometric counterpart to the reference report's "asymmetric transmission" narrative.: discuss coefficient signs, magnitudes, and statistical significance in relation to the theoretical expectations and the 2026 Gulf Crisis context.*

## 14.3 Monetary Policy & Exchange Rate Responses

The coefficients on the policy rate and NEER variables carry important policy implications. A significant negative coefficient on the policy rate for the CPI model suggests that central bank tightening was or would be effective in containing inflation pass-through. Countries where the NEER depreciated sharply (notably Turkey) would be expected to show amplified inflationary effects, consistent with the dual-channel transmission (oil price + currency weakness) documented in the literature.

**MONETARY POLICY & EXCHANGE RATE DISCUSSION**

*The monetary and exchange rate channels are empirically distinct and policy-relevant. In the CPI model, the NEER coefficient of −1.07 (RE) and −1.26 (POLS) implies that a 1% currency appreciation reduces log-CPI by more than 1 percentage point, making the exchange rate the single most powerful near-term anti-inflationary instrument available to central banks with flexible exchange rate regimes. Countries that defended their currencies (notably Germany and France within the Eurozone) experienced materially lower pass-through than Turkey, where the NEER depreciated by over two-thirds over the sample period, amplifying the oil shock via a dual-channel mechanism: higher oil prices in USD, then converted at a weaker domestic currency. The positive and significant policy rate coefficient in the CPI models reflects endogeneity: tightening was a response to inflation, not a cause. This implies that standard OLS-based estimates of monetary policy effectiveness are upward-biased, and that the true anti-inflationary benefit of rate hikes operates with longer lags than the monthly panel can identify. The IPI model's strongly negative policy rate coefficient (−1.82*** in FE) confirms that contractionary monetary policy does suppress industrial activity, creating the classic output-inflation trade-off that policymakers face in the aftermath of a supply-side shock.: discuss coefficient signs, magnitudes, and statistical significance in relation to the theoretical expectations and the 2026 Gulf Crisis context.*

# 15 Conclusion

This project has applied panel data econometrics to quantify the macroeconomic impact of the 2026 Gulf Crisis across a sample of eight economically and geographically diverse countries. Using monthly data from January 2023 through December 2025, three specifications, Pooled OLS, Fixed Effects, and Random Effects, were estimated for both inflation (CPI) and industrial production (IPI) outcomes.

The methodological progression from Pooled OLS to Fixed Effects is not merely a statistical formality. It reflects a substantive recognition that countries differ in deep structural ways: energy dependence, monetary policy credibility, and industrial composition that cannot be captured by observed covariates alone. The diagnostic tests guide the choice of preferred specification, and the subgroup analysis unpacks the heterogeneity that full-sample estimates average over.

The ARIMA forecasts extend the analysis into the post-sample period, providing a quantitative basis for assessing whether the shock appears to be tracking the reference report's moderate or severe scenarios. Together, the econometric and forecasting results offer a rigorous, data-driven complement to the qualitative scenario analysis in the underlying report.

> *The central finding that the oil shock generates a measurable stagflationary impulse, asymmetric across country groups, and partially amplified by exchange rate and monetary policy channels has direct relevance for policymakers navigating the economic aftermath of the 2026 Gulf Crisis.*

### Limitations:

-Several caveats deserve acknowledgment. The panel spans only 42 months, limiting the precision of time-series-based inference.

-The policy rate variable is measured at the end-of-period and may not fully capture intra-month monetary policy responses. Turkey's hyperinflationary dynamics create outlier leverage that, while structurally informative, may affect pooled estimates.

-Future work could extend the sample, explore non-linear threshold effects, and incorporate commodity-specific trade weights to refine the oil exposure measure.

# 16 References

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